

DIPANKAR SAHA

Senior Backend / Full Stack Engineer | AI & Distributed System

React | Next.js | TypeScript | Node.js | NestJS | Microservices | AWS | Kubernetes | AI/LLMs

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Portfolio: dipankarsaha.vercel.app | GitHub: github.com/dip7501686040 | LinkedIn:

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 6 years of experience designing, developing, and deploying scalable enterprise applications, backend services, and cloud-native microservices. Deep expertise in Node.js, TypeScript, NestJS, React, and Next.js, with strong command of REST API design, database engineering, and CI/CD automation. Skilled in building AI-powered applications using OpenAI APIs, with a proven track record in system design, technical architecture, and mentoring engineering teams to deliver secure, production-ready software.

CORE SKILLS

Backend: Node.js, NestJS, TypeScript, REST APIs, GraphQL, gRPC, JWT, OAuth2, Authentication & Authorization

Frontend: React.js, Next.js, JavaScript (ES6+), Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma, Query Optimization & Indexing

Messaging & Events: RabbitMQ, Event-Driven Architecture, Microservices Design

AI & LLM: OpenAI API Integration, Generative AI, Prompt Engineering, LLM-based Workflow Design

Cloud & DevOps: AWS (EC2, S3, EKS), Docker, Kubernetes, Nginx, GitHub Actions, Jenkins, CI/CD

Observability: OpenTelemetry, Prometheus, Grafana, Jaeger, Loki

WORK EXPERIENCE

Featured Project — AI-Powered Event-Driven Notification Platform

Designed and developed a cloud-native notification platform using event-driven microservices and RabbitMQ for asynchronous processing across independent services.

- Architected the end-to-end flow from event creation to rule evaluation, AI processing, and multi-channel notification delivery using RabbitMQ.
- Used gRPC for synchronous inter-service communication and RabbitMQ for asynchronous, event-driven communication across microservices.
- Built an AI-assisted decision engine for intelligent notification processing and routing.
- Implemented authentication, authorization, observability, caching, and production-oriented failure handling including dead-letter exchanges.
- Designed HLD/LLD and demonstrated backend event processing, RabbitMQ messaging, Kubernetes deployment, and OpenTelemetry/Loki/Jaeger observability.

Technologies: Node.js | TypeScript | NestJS | gRPC | RabbitMQ | PostgreSQL | Redis | OpenAI | Docker | Kubernetes | OpenTelemetry

GitHub: github.com/dip7501686040/ai-notification-system | Demo: dipankarsaha.vercel.app/#projects

Senior Software Engineer

Veztraa Solutions (Remote) | March 2023 – Present

- Designed and developed enterprise-scale SaaS applications using React, Next.js, TypeScript, Node.js, and NestJS.
- Led architecture discussions for scalable cloud-native applications following microservices architecture.
- Designed modular backend services with REST APIs and event-driven messaging.

- Built reusable frontend components and improved maintainability through component-driven architecture.
- Implemented secure authentication and authorization using JWT and Role-Based Access Control (RBAC).
- Optimized MongoDB aggregation pipelines and MySQL queries, improving query performance through indexing and schema tuning.
- Containerized applications using Docker and deployed workloads to Kubernetes clusters on AWS.
- Built automated CI/CD pipelines using GitHub Actions and Jenkins.
- Participated in architecture reviews, sprint planning, technical design discussions, and code reviews.
- Mentored junior engineers and promoted engineering best practices across the team.

Technologies: React | Next.js | TypeScript | Node.js | NestJS | Docker | Kubernetes | MySQL | MongoDB | RabbitMQ | AWS | GitHub Actions | Jenkins

Software Engineer

Tata Consultancy Services (TCS) (Kolkata) | April 2021 – Feb 2023

- Developed enterprise insurance applications for Zurich Insurance using Angular and Node.js.
- Built scalable REST APIs and frontend modules supporting high-volume business workflows.
- Worked closely with distributed teams following Agile/Scrum methodology.
- Fixed production issues while maintaining high application availability.
- Participated in code reviews and delivered high-quality software following enterprise standards.
- Improved application performance and reduced production defects through proactive optimization.

Technologies: Angular | Node.js | JavaScript | REST APIs | MySQL | Git

Software Engineer

Asparrowtech (Indore) | June 2020 – March 2021

- Developed responsive web applications using JavaScript frameworks and Node.js.
- Built backend APIs and integrated third-party services.
- Worked on database design, application enhancements, and bug fixes.
- Collaborated with clients to deliver custom software solutions.

EDUCATION

Bachelor of Technology (Information Technology) — Siliguri Institute of Technology | 2016 – 2020 | CGPA: 7.70

PROFESSIONAL STRENGTHS

Backend Engineering, Full Stack Architecture, Event-Driven Architecture, AI-Enabled Product Development, Microservices Design, Cloud-Native Engineering, System Design, Database Design & Optimization, Secure API Development, Performance Optimization, Technical Leadership, Mentoring Engineers, Scalable SaaS Platforms, Agile Delivery, Good Communication

ADDITIONAL INFORMATION

Languages: English, Hindi, Bengali | Work Authorization: India | Open to Remote Opportunities | Open to Relocation